Senda LAATAOUI

AI Researcher | Deep Learning, NLP and Computer Vision

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Summary

I'm an AI engineer with a research Master's in Deep Learning, and I'm looking for a PhD in artificial intelligence. My work so far has focused on explainable AI for medical data, from Parkinson's diagnosis at BASIRA Lab (Imperial College London) to cardiac MRI segmentation, and on NLP for scientific text. I like projects where models have to be understandable and useful to people outside machine learning, and I'm looking for an interdisciplinary team where I can grow as a researcher.

Research interests: Explainable AI for medical imaging · NLP for scientific text · Trustworthy and interpretable deep learning

Research and Teaching Experience

Maghreb Institute of Economics and Technology (IMSET) — Sousse

Workshop Designer and Facilitator, "Inside the Black Box: Can We Trust AI?" — May 2026

- Designed and delivered the workshop for 50+ students, centered on the LIME explainability method.
- Used examples of algorithmic bias, model opacity and AI ethics to lead discussions on responsible AI deployment.

Maghreb Institute of Economics and Technology (IMSET) — Sousse

Artificial Intelligence Instructor — March-June 2026

- Taught AI fundamentals to more than 80 students.
- Delivered training in Machine Learning, NLP, computer vision and Deep Learning.
- Supervised practical lessons on classification, text generation and image processing.

Brain and Signal Research and Analysis Laboratory (BASIRA Lab) — London

Deep Learning Intern — March-December 2025

- Designed a **prototype-based interpretable multi-task U-Net** for spiral-trace segmentation and 4-stage Parkinson's severity classification.
- Built and preprocessed the annotated spiral dataset with ground-truth masks and clinical severity labels.
- Delivered built-in explainability via learned prototypes and prototype-to-class contribution maps showing which drawing traits drove each prediction.
- Achieved **Dice 0.87, cLDice 0.86, and 87.3% 4-class accuracy**; multi-task training outperformed single-task baselines.
- Collaborated with a multidisciplinary team spanning machine learning and clinical neurology.

Medical Imaging and Technology Laboratory (Lab-TIM) — Tunisia

Deep Learning Intern — July-August 2024

- Developed and optimized STUNet for cardiac MRI segmentation.
- Improved accuracy by 8% over U-Net, achieving a Dice score of 0.91.
- Reduced processing time by 40% through pipeline automation.
- Achieved inference in under two seconds per examination.

Education

Polytechnic School of Monastir — Engineering Degree

Artificial Intelligence — 2024-2026

Ranked 1st in my class throughout the engineering program [Average: 17/20; highest honors]

Final-year project: Named Entity Recognition in a corpus of heliophysics research papers

Higher Institute of Computer Science and Mathematics of Monastir — Bachelor's and Research MSC degrees

Software Engineering, Deep Learning specialization — 2020-2025

Ranked 2nd in my Master's class [Average: 14.5/20; honors]

Thesis: Explainable Medical Data Analysis for Automated Diagnosis (BASIRA Lab, Imperial College London)

Community and Academic Engagement

Google Developer Groups (GDG) 2026-Present
Organizer and Overall Coordinator, DevFest 2026. The event is scheduled for October 31, 2026.

MICCAI (Medical Image Computing and Computer-Assisted Intervention Society) 2025-Present
Participate in the society's activities and medical imaging community.

ARSII (Association for Scientific Research and Innovation in Computer Science) 2024-Present
Participate in organizing events and in association activities.

Research Projects

Named Entity Recognition Pipeline for Heliophysics Literature February-September 2026

- Built and annotated a corpus of **102 articles** with **34,888 entity annotations** and **672 figure captions** annotated across three attributes (figure type, measurement unit, observable).
- Developed a **Docling** PDF-extraction pipeline handling two-column layouts and linking captions to figures.
- Fine-tuned **ModernBERT** for token classification: **macro F1 0.682** vs **0.300** (rule-based) and **0.652** (SciBERT).
- Fine-tuned **FLAN-T5-Large**: **72.4%** figure-type accuracy and **0.715 strict F1** for measurement units, with **no invalid outputs** on the test set.
- Technologies: Python, PyTorch, Hugging Face Transformers, Docling, Label Studio.

Mask Detector, End-of-Year Project (PFA) February-May 2025

- Developed a local, real-time face detection and tracking system.
- Selected and optimized YOLOv8, achieving 89.3% mAP50 and 45 FPS.
- Developed a Python/Flask back end for video processing and an Angular front end.
- Integrated MySQL, WebSockets and Docker, achieving latency below 100 ms and tracking more than 80 faces.

Skills

Artificial Intelligence and Research
Deep Learning, Machine Learning, NLP, Computer vision, Explainable AI, Python, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, CUDA

Research and Collaboration Tools
Git, GitHub, Jupyter, MLflow, Label Studio, Docling, Docker

Certifications

Artificial Intelligence Certification Dec. 2025
Pearson

TOEIC - C1 English Level Mar. 2025
Educational Testing Service

Python - Next Generation Jan. 2026
Pearson

Languages

French
C1 (Advanced)

English
C1 (Advanced)

Italian
B1 (Intermediate)